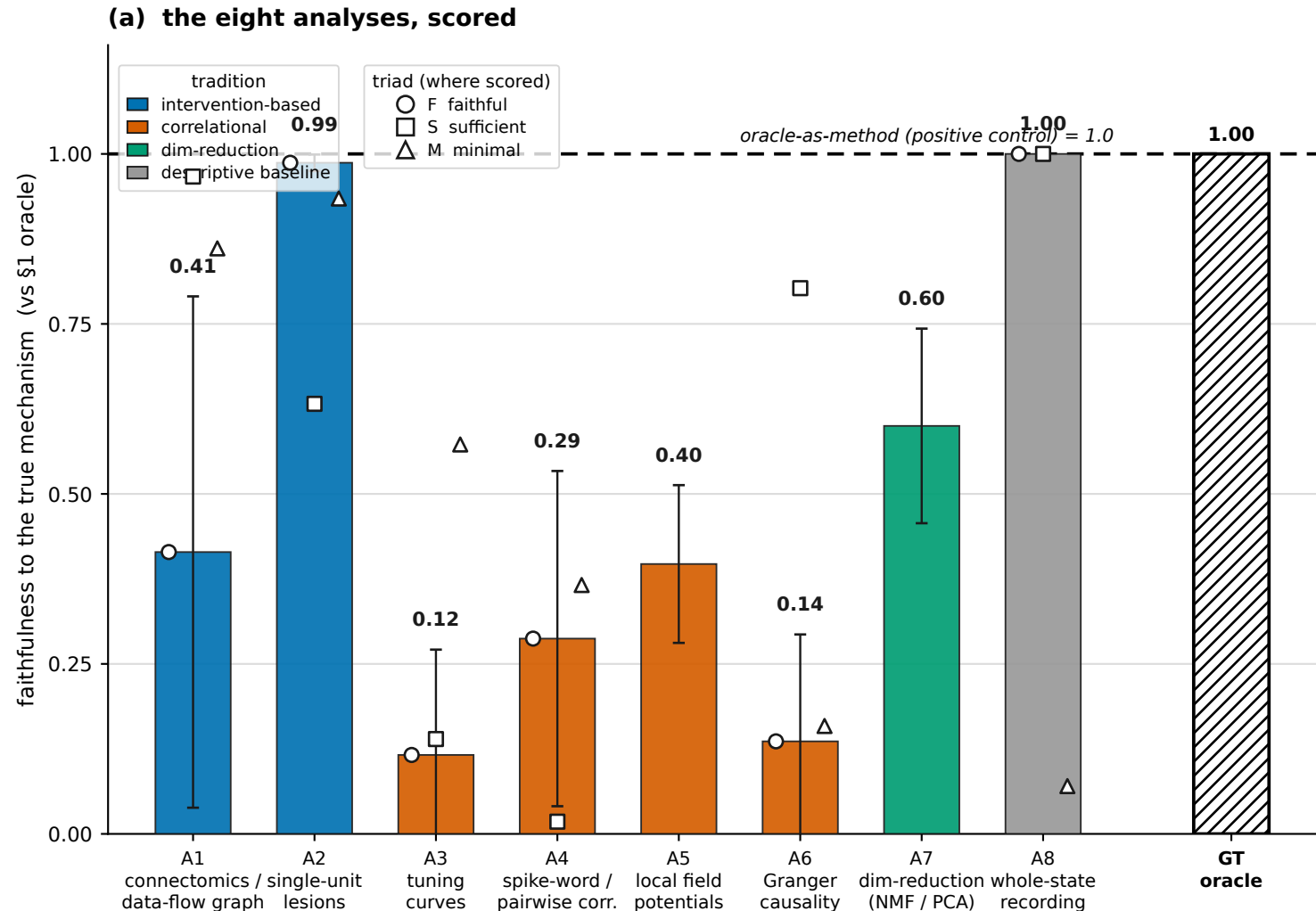


The neuroscience battery on a known machine — scored against ground truth

Jonas & Kording's battery, quantified: classical analyses find rich structure yet score low in faithfulness to the true register-transfer mechanism; the oracle-as-method control = 1.0.



(b) key findings

A6 **Granger F = 0**

Pong: false-edge rate = 1.00,

missed-edge = 1.00 → triad F = 0.00.

Temporal precedence ≠ causation: on a

A3 **spurious tuning**

clock-locked machine Granger infers spurious bidirectional edges everywhere.

Pong spurious-tuning rate = 1.00;

Space Invaders = 0.67, Q*bert = 0.53.

Strongly-tuned units whose tuning does

A2 **p = 0.99, blind to interactions**

NOT match their true causal role — the classic tuning-curve trap, quantified.

Single-unit lesion importance p = 0.99

vs the true role — yet single lesions miss 40% of interactions (Space Inv.).

A7 **NMF > PCA (both mediocre)**

super-additive pair Δ up to 1044 invisible to one-unit-at-a-time ablation.

NMF matched-component fraction = 0.60

vs PCA = 0.60 mean (NMF best 0.8).

Non-negativity aligns components with

Data (no re-run): per-method faithfulness + triad F/S/M from tools/xai_study/compare/out/leaderboard.json (E6-1); annotation numbers from tools/xai_study/phaseA_kording/out/A{1..8}_*.json. 6 core games, whiskers = 95% CI.